Proof by Mathematical Induction — Problem Set

Warm-up

1. Use mathematical induction to prove $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$ for $n \geq 1$.
2. Prove that $5^n - 1$ is divisible by 4 for $n \geq 1$.
3. Use mathematical induction to prove $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for $n \geq 1$.
4. Prove that $3^n - 1$ is an even integer for all $n \geq 1$.
5. Show that $1 + 4 + 7 + \cdots + (3n-2) = \frac{n(3n-1)}{2}$ holds for the base case $n = 1$.
6. If $P(k) = 10^k - 1$ is divisible by 9, prove $P(k+1) = 10^{k+1} - 1$ is also divisible by 9.
7. Prove $2^n > n$ for $n = 1, 2, 3$.
8. Show that $\sum_{i=1}^{n} r^{i-1} = \frac{r^n - 1}{r-1}$ for the case $n = 1$.
9. Prove that $n^2 + n$ is always even for any positive integer n using induction.

Skill-building

1. Prove by mathematical induction that $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for $n \geq 1$.
2. Use mathematical induction to show that $4^n + 2$ is divisible by 3 for all integers $n \geq 1$.
3. Prove that $1 \times 2 + 2 \times 3 + 3 \times 4 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}$ for $n \geq 1$.
4. Prove that $8^n - 3^n$ is divisible by 5 for all positive integers n.
5. Use induction to prove $\frac{1}{1\times 2} + \frac{1}{2\times 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ for $n \geq 1$.
6. Prove that $x^n - y^n$ is divisible by $(x - y)$ for $n \geq 1$ (where $x \neq y$).
7. Prove by induction that $2^{3n} - 1$ is a multiple of 7 for all $n \geq 1$.
8. Prove that $(1 + x)^n \geq 1 + nx$ for $x > -1$ and $n \geq 1$.
9. Show that $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$ for all positive integers n.
10. Prove by mathematical induction that $10^{n+1} + 3 \cdot 10^n + 5$ is divisible by 9 for $n \geq 0$.

Easier Exam Questions

1. (Newington 2024 Q12a) Use mathematical induction to prove that **3**

$$2^3 + 4^3 + 6^3 + \cdots + (2n)^3 = 2n^2(n+1)^2$$

2. (Manly 2020 Q12a) Use mathematical induction to prove that $7^{2n-1} + 5$ is divisible by 12 for all integers $n \geq 1$. **3**

3. (Sydney Grammar 2020 Q12d) Use mathematical induction to prove that, for positive integers n: **3**

$$1 + 3 + 9 + \cdots + 3^{n-1} = \frac{1}{2}(3^n - 1).$$

4. (Gosford 2019 Q13b) Use the principle of Mathematical Induction to prove that $7^n - 3^n$ is a multiple of 4 for all positive integral values of n. **3**

5. (Fort Street 2021 Q2a) Consider the claim: $1 + 2 + 3 + \cdots + n = \frac{1}{2}\left(n + \frac{1}{2}\right)^2$ is true for all positive integer n.

   (i) Assuming the claim is true for $n = k$, where k is a positive integer, then prove it is true for $n = k + 1$. **2**

   (ii) Prove the claim to be false for $n = 1$? **1**

   (iii) Hence justify why the original statement is false. **1**

6. (Blacktown Boys 2020 Q13a) Using mathematical induction, prove the following is true for all positive integers n **3**

$$1 + 6 + 11 + \cdots + (5n - 4) = \frac{1}{2}n(5n - 3)$$

Harder Exam Questions

1. (Hornsby Girls 2023 Q11e)

   (i) For positive integers n and r, with $r < n$, show that **2**

$$^nC_r + {^nC_{r+1}} = {^{n+1}C_{r+1}}$$

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   (ii) Hence, prove, by Mathematical Induction, that **3**

$$\sum_{j=3}^{n} {^{j-1}C_2} = {^nC_3}$$

   for all integers $n \geq 3$.

2. (Sydney Tech 2020 Q13c) Use mathematical induction to prove that $7^{2n} + 7^n + 4$ is divisible by 6 for integers $n \geq 0$. **4**

3. (Normanhurst Boys 2025 Q13a)

   (i) Show that $\frac{1}{(n+2)!} - \frac{n+1}{(n+3)!} = \frac{2}{(n+3)!}$, where n is an integer. **2**

   (ii) Prove by mathematical induction that, for all positive integers n, **3**

$$\frac{1 \times 2}{3!} + \frac{2 \times 2^2}{4!} + \frac{3 \times 2^3}{5!} + \cdots + \frac{n \times 2^n}{(n+2)!} = 1 - \frac{2^{n+1}}{(n+2)!}$$

4. (Normanhurst Boys 2023 Q13a) Prove by mathematical induction that for all positive integer values of n, **3**

$$(1^2 \times 2^1) + (2^2 \times 2^2) + (3^2 \times 2^3) + \cdots + (n^2 \times 2^n) = (n^2 - 2n + 3) \times 2^{n+1} - 6$$

5. (Blacktown Boys 2025 Q12b) Prove by mathematical induction that $3^{2n} + 16n - 1$ is divisible by 8 for all positive integers n. **3**

6. (Knox 2022 Q12c) Use the principle of mathematical induction to show that for all integers $n \geq 1$, **3**

$$3 \times 5^{2n+1} + 2^{3n+1} \text{ is divisible by 17.}$$

7. (Blacktown Boys 2024 Q12f)
For all integers $n \geq 1$, use mathematical induction to prove that: **3**

$$\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \cdots + \frac{n+2}{n(n+1) \times 2^n} = 1 - \frac{1}{(n+1) \times 2^n}$$

8. (Blacktown Boys 2022 Q14a)

   (i) Prove that $n + 1$ is a factor of $P(n) = 4n^3 + 18n^2 + 23n + 9$. **1**

   (ii) Hence, use mathematical induction to prove that for all integers $n \geq 1$, **3**

$$1 \times 3 + 3 \times 5 + 5 \times 7 + \cdots + (2n+1)(2n-1) = \frac{n(4n^2 + 6n - 1)}{3}$$